

Chhattisgarh. The Energy and Resources Institute has launched a research project on jatropha plant. While India is the fourth largest producer of ethanol, thanks to its sugarcane output, much of this goes into the petrochemical and liquor industries. What prevents ethanol from being used as a bio-fuel is policy, coupled with unstable production. Although a 5 per cent ethanol blending with petrol was mandated in nine states and four Union Territories in January 2003, the project never quite took off as oil companies could only buy less than half their requirement of ethanol from sugar factories. This, with the price—higher than the import price of petrol—fixed by the government for purchase by oil companies had them backing out. Moreover, levies, which range from import duties in certain states to export fees in others along with excise cess, also acted as constraints.

However, this is not to take away from the potential of investing in bio-fuels in India, which saw increased private investment in 2006. The experiences of clean-tech companies like Suzlon and Praj have been more than encouraging. Citigroup Venture Capital International, which invested in Suzlon in 2004, has made a killing as its original investment of Rs 100 crore has grown to Rs 2,600 crore now. Similarly, ChrysCapital's Rs 50-crore equity investment in Suzlon is currently worth over 50 times its cost basis. Says Praj Chairman Pramod Chaudhary: "Growing investor interest suggests our technology is going to scale up." Last year, US-based venture capitalist Vinod Khosla, who has backed companies like Google and Amazon, invested Rs 117 crore in Praj, while three global funds together pumped \$20 million (Rs 80 crore) in electric car maker Reva. Meanwhile, investors like Warburg Pincus are eyeing a stake in Moser Baer PV valuing the three-year-old company at \$1.5 billion (Rs 6,000 crore).

Today, many like the Xenitis Group, Ahmedabad's Electrotherm India and bicycle makers Hero and Atlas have made a foray into electric motorbikes.



BANDEEP SINGH

"We have orders and agreement up to \$100 million already in hand."

RATUL PURI, EXECUTIVE DIRECTOR, MOSER BAER

THE SOLAR POWER major plans an installed capacity of 400-500 MW in two-three years.

Auto maker Mahindra & Mahindra (M&M) too has launched a bio-diesel-powered Scorpio and Bolero. The company set up its own bio-diesel plant in 2001. "For every 10 per cent bio-diesel mixed, the particle emissions will reduce by 8-9 per cent and fuel efficiency will go up by 10 per cent," claims Arun Jaura, v-p, engineering and R&D, M&M.

Significantly, unlike the world over, Indian investments have been in late-stage companies or those looking to diversify. "That's because we are busy investing in domestic consumption-led growth," says Arun Natarajan, CEO of Venture Intelligence. Adds Promod Haque, managing partner at Norwest Venture Partners: "There is a two-year lag between the US and India. Web 2.0 is just kicking off now." Analysts also attribute it to the long gestation period of projects. "Biotech was touted as the next big thing after IT. But who has invested in it?" asks Natarajan.

One reason for the hesitation is the

risks that come with clean-tech as its economics and profitability vary with changes in prices and policies. "Our political framework does not give a strong message towards clean tech. Then why will global investors look here?" asks Tanti. Sandeep Murthy, India representative of Kleiner Perkins, says the demand is huge but the market will depend on the official policies that determine pricing.

The question is: will the 2020 goal, to ensure that 20 per cent of the country's energy comes from alternative sources, be achieved? The answer is not yet clear as there are no mandatory Central policies for utilities to buy from renewables. Recently, the Renewable Energy Ministry announced a Rs 25,000-crore roadmap to promote the use of hydrogen in about one million vehicles by 2020. It has sanctioned a project for setting up a hydrogen dispenser at a petrol pump in Delhi. This will enable dispensing of neat hydrogen and CNG blended with hydrogen as fuel for vehicles.

The interest in alternative energy is expected to increase significantly as India, according to the International Energy Agency, will represent one-third of the global energy demand by 2050. The Planning Commission estimates the country will need to increase its primary energy supply by three-four times and its electricity supply by five-seven times to maintain an 8-10 per cent growth up to 2030. As the need for renewable energy becomes imperative, a study by Clean Edge, a research and consulting group, shows that the total market for biomass, pv, wind, energy and fuel cells will grow fourfold from \$40 billion (Rs 1,60,000 crore) in 2005 to \$167 billion (Rs 6,68,000 crore) globally by 2015. And by 2012, adds Ernst & Young (E&Y), India and China will be the most attractive options for renewable energy investors. "There is a huge opportunity for India as investment in cleaner products and technologies can lead to a potent new source of innovation and competitive advantage," says Jonathan Johns, head of clean consulting, E&Y. It means clean tech has a bright future. ■